Role of the Vishniac Magnetic Helicity Flux in Mean-field Galactic Dynamos

A report submitted By

Gayathri K

School of Physical Sciences

National Institute of Science, Education and Research (NISER), Bhubaneswar

As part of

Visiting Student Programme 2025

Indian Institute of Astrophysics (IIA), Bengaluru

Under the Guidance of

Dr. Sharanya Sur

Associate Professor

Indian Institute of Astrophysics (IIA), Bengaluru

Acknowledgement

I would like to express my sincere gratitude to the Indian Institute of Astrophysics (IIA), Bangalore, and the Board of Graduate Studies for organizing the Visiting Students Programme (VSP), which has been a truly enriching experience. I am especially thankful to my master's thesis supervisor, Dr. Luke Chamandy, for his constant support and guidance, and to Dr. Sharanya Sur, my VSP supervisor at IIA and thesis collaborator, for her mentorship and encouragement.

Working with Dr. Sharanya Sur has been an invaluable experience. Our discussions helped me uncover subtle nuances and deepen my understanding of the connections between various theoretical concepts I have been exploring for nearly two years.

I would also like to thank my collaborators, Prof. Kandaswamy Subramanian and Dr. Kishore Gopalakrishnan, for their valuable input and inspiring discussions. Your work laid the foundation for mine, and I consider myself lucky to have learned from you.

A special thanks to Dr. Frederick A. Gent for generously sharing his simulation data, which formed an important part of this project.

Gayathri K

5th Year Integrated M.Sc.

School of Physical Sciences

NISER, Bhubaneswar

Abstract

Magnetic fields are crucial to the evolution of galaxies, but the mechanisms amplifying and sustaining large-scale fields remain unclear. In disc galaxies, the evolution of large-scale magnetic fields is explained through mean-field dynamo theory. This theory focuses on the mean magnetic field ($\overline{\mathbf{B}}$), while the role of random magnetic fields ($\mathbf{b}$) is often overlooked. These smaller-scale fields serve as a helicity reservoir. Without effective fluxes to expel excess helicity, the buildup of magnetic helicity can suppress the α-effect, which is proportional to the mean kinetic helicity density. This process is known as catastrophic quenching. Helicity fluxes can counteract this by redistributing helicity. In this study, we incorporate one such helicity flux, the Vishniac flux, into numerical models for the first time, building on the analytical framework of Gopalakrishnan & Subramanian (2023). Using a numerical dynamical quenching model, we assess the Vishniac flux's role in the evolution of galactic magnetic fields. The results suggest the possibility of a magnetically driven dynamo mechanism under specific conditions, such as strong rotation and super-equipartition small-scale magnetic fields. This study highlights the potential role of helicity fluxes in sustaining large-scale magnetic fields and motivates further investigation into magnetic contributions to dynamo action.

Contents

Acknowledgement **i**

Abstract **ii**

Page

1 Introduction 1

2 Theoretical Background 2

2.1 The Diffusive Helicity Flux 3

2.2 The New Vishniac Flux 5

3 The new induction equation 6

4 The model 6

4.1 Assumptions 6

4.2 Numerical Methods 7

4.3 Initial and Boundary Conditions 7

4.4 Implementation 7

5 Results and Analysis 8

5.1 The Vishniac Flux in Mean-Field Dynamos: Aid or Hindrance? 8

5.2 The Vishniac flux action without α_k driving the dynamo 9

6 Conclusions 11

1 Introduction

Magnetic fields are key in many astrophysical systems and processes, from star formation to cosmic ray propagation in the interstellar medium (ISM). This work focuses on magnetic fields in disc galaxies, where observations reveal large-scale, coherent fields with typical strengths of around 10μG. A natural first question is: where do these magnetic fields come from? Are they primordial, i.e., fossil relics from the early universe, or is there an ongoing mechanism that continuously regenerates them?

To answer this, we turn to the nature of the plasmas that host these fields. These plasmas are highly turbulent. A defining feature of turbulence is the cascade of energy to smaller scales. Thus, a primordial magnetic field would struggle to survive over long timescales in high turbulence environments, as magnetic energy drains to smaller scales; ohmic dissipation is both dominant and efficient. This suggests that some mechanisms must actively sustain and even amplify these fields. That mechanism is the hydromagnetic dynamo, which refers to the process by which kinetic energy is spontaneously converted into magnetic energy.

The evolution of magnetic fields can be very well described by the equations of single-fluid magnetohydrodynamics (MHD) in most astrophysical plasmas. In this formalism, the plasma behaves as a conducting fluid influenced by fluid and electromagnetic forces. However, solving the full MHD equations in turbulent regimes presents significant computational challenges due to the vast range of spatial and temporal scales involved. Thus, we rely on simplified models to study these fields. A well-accepted model is the mean field dynamo theory, where fields are decomposed into mean (large-scale) and fluctuating (small-scale) components, and we focus on mean-field growth. This theory relies on turbulent flows breaking mirror symmetry, enabling magnetic fields to grow at scales larger than the turbulence[1][8].

Kinetic helicity is one of the most natural measures of the lack of reflectional symmetry and it arises naturally in rotating systems. The resulting large-scale magnetic fields are characterized by magnetic helicity, which measures the twist, writhe, and linkage of field lines. Kinetic and magnetic helicity are central to understanding magnetic field amplification and evolution[2]. Here, we adopt the $\alpha\Omega$ mean field dynamo model. The α-effect converts toroidal magnetic fields into poloidal fields through the action of kinetic helicity, while the Ω-effect converts poloidal fields back into toroidal fields, forming the basis of the $\alpha\Omega$-dynamo model.

However, the amplification of large-scale magnetic fields in mean-field dynamos is limited. As large-scale fields grow, small-scale magnetic helicity of opposite sign to that of the large-scale magnetic helicity is generated to conserve total magnetic helicity. As this small-scale magnetic helicity accumulates, it reduces dynamo efficiency by suppressing the α-effect. Mechanisms to remove or redistribute small-scale helicity are crucial to maintaining dynamo activity. While resistive dissipation is too slow, processes such as large-scale outflows may expel helicity. However, in many astrophysical systems, such large outflows may not be present, which calls for alternative mechanisms to facilitate helicity transfer. Thus, the standard dynamo models may not fully capture the mechanisms driving large-scale magnetic field growth.

Vishniac (2012) [10] proposed a new effect arising from higher-order terms, known as the Vishniac flux, which is a magnetic helicity flux. The Vishniac flux arises from gradients in the random magnetic and kinetic energies along the direction of large-scale rotation. As a result, it can operate in any rotating, stratified system. This project aims to incorporate the Vishniac flux into the traditional mean-field dynamo model and assess its impact on the evolution of galactic magnetic fields. The analytic expression for the Vishniac flux, derived by Gopalakrishnan & Subramanian (2023), is integrated into a dynamical quenching model developed by Chamandy et al. (2014) [4]. Our model simultaneously solves the mean-field induction equation and the

helicity evolution equation.

Unlike traditional models that primarily rely on kinetic energy-specifically the kinetic α-effect and helicity,as the main drivers of the dynamo, our model suggests that magnetic energy, in combination with large-scale rotation or shear, could be sufficient to sustain the dynamo. However, this is only true when the small-scale field strength exceeds the turbulent kinetic energy, i.e., in a super-equipartition state where small-scale fields dominate. This perspective challenges some foundational aspects of dynamo theory and aligns with our goal of providing a more comprehensive explanation for the observed magnetic field structures in galaxies.

2 Theoretical Background

The evolution of the magnetic field in a plasma is governed by the induction equation:

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{V} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}, \tag{1}$$

where $\eta = c^2/(4\pi\sigma)$ is the magnetic diffusivity. The first term in Eq. (1) is the induction term, representing magnetic field generation by plasma motion, and the second term is the diffusion term, representing field decay due to finite conductivity.

Using Reynolds averaging, the velocity and magnetic fields are decomposed as:

$$\mathbf{V} = \overline{\mathbf{V}} + \mathbf{v}, \quad \mathbf{B} = \overline{\mathbf{B}} + \mathbf{b}, \tag{2}$$

where $\overline{\mathbf{F}}$ is the mean and $\mathbf{f}$ the fluctuating component for a field $\mathbf{F}$ with $\overline{\mathbf{v}} = \overline{\mathbf{b}} = 0$.

Applying this to Eq. (1), the mean-field induction equation becomes:

$$\frac{\partial \overline{\mathbf{B}}}{\partial t} = \nabla \times \left(\overline{\mathbf{V}} \times \overline{\mathbf{B}} + \boldsymbol{\mathcal{E}} - \eta \nabla \times \overline{\mathbf{B}} \right), \tag{3}$$

where $\boldsymbol{\mathcal{E}} = \overline{\mathbf{v} \times \mathbf{b}}$ is the mean electromotive force (EMF).

The mean EMF is expressed as:

$$\boldsymbol{\mathcal{E}} = \alpha \overline{\mathbf{B}} - \eta_t \nabla \times \overline{\mathbf{B}}, \tag{4}$$

where α represents helicity-driven amplification of the magnetic field (the α-effect) and η_t accounts for turbulent diffusivity, which opposes field growth.The α-effect works by generating electric currents through helical turbulent motions, which interact with the toroidal magnetic field. These currents create a force that "twists" the toroidal field into a poloidal field. Essentially, the motion of the fluid moves the magnetic field lines in such a way that the circular toroidal field becomes a looped poloidal field, transferring the magnetic energy from one configuration to the other. This is how the α-effect converts a toroidal field into a poloidal field. In disc galaxies, the mean velocity field $\overline{\mathbf{V}}$ is dominated by differential rotation. This rotation induces the Ω-effect. This effect involves the stretching of magnetic field lines by differential rotation. It converts poloidal fields into toroidal fields, enabling a regenerative cycle when combined with the α effect. Together, the α and Ω effects form the $\alpha\Omega$-dynamo, supporting exponential growth of the mean magnetic field ($\overline{\mathbf{B}}$) if they overcome turbulent diffusion.

The α-effect in Equation (4) is represented as the sum of kinetic and magnetic contributions:

$$\alpha = \alpha_k + \alpha_m, \tag{5}$$

where:

$$\alpha_k = -\frac{\tau}{3}\overline{\mathbf{u}\cdot(\nabla\times\mathbf{u})} \quad \text{is the kinetic part related to the mean helicity of the random flow,} \tag{6}$$

$$\alpha_m = \frac{\tau}{12\pi\rho}\overline{\mathbf{b}\cdot(\nabla\times\mathbf{b})} \quad \text{is the magnetic contribution.} \tag{7}$$

Here, ρ is the gas density, τ is the correlation time of the random flow, and $\mathbf{u}$ and $\mathbf{b}$ represent the small-scale velocity and magnetic field, respectively.

In the early stages of dynamo action, the magnetic field is weak and does not significantly influence the fluid motions. The growth of the mean magnetic field is exponential, driven by induction effects, with turbulent diffusion attempting to dissipate it. This is the kinematic regime of dynamo action. As the magnetic field grows, it starts to exert a back-reaction on the flow via the Lorentz force. This alters the dynamics of the random velocity field, reducing the effectiveness of the kinetic α_k-effect and turbulent diffusivity (β). The system enters a non-linear regime where the magnetic helicity and magnetic α_m-effect become significant, introducing feedback mechanisms. α_m and α_k have opposite signs, thus as $\overline{\mathbf{b}\cdot(\nabla\times\mathbf{b})}$ is amplified, the magnitude of the total α-effect decreases, saturating the growth of the mean magnetic field.

The dynamo non-linearity is governed by the following equation for α_m-effect:

$$\frac{\partial\alpha_m}{\partial t} = -\frac{2\eta_t}{l^2 B_{\text{eq}}^2}\boldsymbol{\mathcal{E}}\cdot\overline{\mathbf{B}} - \nabla\cdot\boldsymbol{\mathcal{F}}, \tag{8}$$

where l is the correlation length of the turbulence. $B_{\text{eq}} = \sqrt{4\pi\rho}\,u$ is the characteristic magnetic field strength corresponding to energy equipartition with turbulence(this is B_0 at midplane) and $\boldsymbol{\mathcal{F}}$ is the flux density of α_m, related to the flux density of the small-scale magnetic helicity density.

There are many possible contributions to $\boldsymbol{\mathcal{F}}$. In general, a flux of the form

$$\boldsymbol{\mathcal{F}} = \boldsymbol{\mathcal{F}}^{NV} + \boldsymbol{\mathcal{F}}^{A} + \boldsymbol{\mathcal{F}}^{D} \tag{9}$$

is considered. In this project, we are not including the advective flux; thus, we have the diffusive flux and the Vishniac flux for helicity transfer.

2.1 The Diffusive Helicity Flux

In earlier work,the diffusive helicity flux was modeled heuristically, with the following expression for the flux density [4]:

$$\boldsymbol{\mathcal{F}}^{D} = -\kappa\nabla\alpha_m \tag{10}$$

with $\kappa \approx 0.3\eta$, obtained from numerical simulations.
In Gopalakrishnan & Subramanian (2023), the flux density was derived more rigorously by considering higher-order terms in the magnetohydrodynamic equations. This newer form of the diffusive helicity flux is based on the physical processes underlying turbulence and the evolution of magnetic helicity. The form derived is:

$$\boldsymbol{\mathcal{F}}^{D} = -\frac{7\tau}{27}\left(\langle\mathbf{v}^2\rangle + \frac{\langle\mathbf{b}^2\rangle}{4\pi\rho}\right)\nabla h^b \tag{11}$$

Here, ρ is the gas density, τ is the correlation time of the random flow, and $\mathbf{v}$ and $\mathbf{b}$ represent the small-scale velocity and magnetic field, respectively. Following Sur et al. (2007), we define the magnetic helicity density as:

$$\frac{\partial \chi}{\partial t} + \nabla \cdot F = -2E \cdot B - 2\eta j \cdot b \tag{12}$$

$$\alpha_m = \frac{1}{3}\tau \frac{1}{\lambda^2}\frac{\chi}{\rho}\left(\frac{1}{4\pi}\right) \tag{13}$$

Assuming $\chi = h^b$ and $\lambda = v\tau$, we get:

$$h^b = \frac{3\rho\lambda^2}{\tau}\alpha_m(4\pi) = \frac{3\rho\tau^2 v^2}{\tau}\alpha_m(4\pi) = 3\rho\tau v^2 \alpha_m(4\pi) \tag{14}$$

To proceed with the analysis, we considered the following turbulence profiles:

$$\text{Turbulent velocity} \quad v^2(z) = v_0^2 \exp\left(\frac{Az^2}{h^2}\right), \qquad \text{Turbulent magnetic field} \quad b^2(z) = v_0^2 \exp\left(-\frac{z^2}{h^2}\right), \tag{15}$$

where v_0 and b_0 are characteristic turbulent velocity amplitudes and h is the disc scale height. The A in the exponential of this velocity profile is used to produce different cases of stratification in our models.

For consistency with the assumptions made in the theoretical derivation of the fluxes[6], we keep the mass density ρ constant throughout the simulations. This assumption simplifies the analysis and allows us to focus on the behavior of the turbulent velocity and magnetic fields, which are the primary variables driving the helicity flux we are interested in.
To express the ratio of magnetic to kinetic turbulent energies, we define the parameter:

$$\xi(z) = \frac{b^2(z)}{4\pi\rho v^2(z)}. \tag{16}$$

This is written as:

$$\xi(z) = \tilde{\xi}(z)\,\xi_0, \tag{17}$$

where $\xi_0 = \frac{b_0^2}{4\pi\rho v_0^2}$ is a constant. Refer to Table 1 for how $\xi(z)$ behaved with different values of A we considered in the study.

Now we derive the form of diffusive flux given in equation **??** for the turbulence profiles, we get

Table 1: List of models

Model	A	ξ
Case 1	1	decreases with z
Case 2	-1	constant
Case 3	-2	increases with z

the divergence of flux to be:

$$-\nabla\cdot\boldsymbol{\mathcal{F}}^{D} = \frac{7\eta}{9}(1+\xi)\left[\nabla^2\alpha_m + \frac{4A^2z^2}{h^4}\alpha_m + \frac{2A}{h^2}\alpha_m + \frac{4Az}{h^2}\nabla\alpha_m\cdot\hat{z}\right] + \frac{7\eta}{9}\left(\frac{2z}{h^2}\right)(A-\xi)\left(\frac{\partial\alpha_m}{\partial z} + \frac{2Az}{h^2}\alpha_m\right)$$

2.2 The New Vishniac Flux

$\boldsymbol{\mathcal{F}}^{NV}$ is the New Vishniac flux, not to be confused with the Vishniac-Cho flux(Vishniac & Cho 2001). Henceforth, the term "Vishniac flux" will refer to the New Vishniac flux, this flux is the main focus of the project. This is derived by Gopalakrishnan & Subramanian (2023) and is expressed as:

$$\boldsymbol{\mathcal{F}}^{NV} = (\nabla\times\overline{\mathbf{V}})\cdot\left[C_1\frac{\tau}{8\pi\rho}(\langle\mathbf{b}^2\rangle)^2 + C_2\tau\langle\mathbf{v}^2\rangle\langle\mathbf{b}^2\rangle + C_4\lambda^2\langle\mathbf{b}^2\rangle\right]\hat{z} + \tau(C_3-C_2)(\overline{\mathbf{V}}\times\langle\mathbf{v}^2\rangle\nabla\langle\mathbf{b}^2\rangle) \tag{18}$$

This expression simplifies under the assumption that all turbulent correlators predominantly vary in a single spatial direction, which is physically justified in systems like disk galaxies (with vertical stratification). Thus when we take a divergence of this flux, the second term will vanish. Thus, we have:

$$-\nabla\cdot\boldsymbol{\mathcal{F}}^{NV} = -(\nabla\times\overline{\mathbf{V}})\cdot\left[C_1\frac{\langle b^2\rangle}{4\pi\rho}\tau\nabla\langle b^2\rangle + C_2\langle b^2\rangle\tau\nabla\langle v^2\rangle + C_3\langle v^2\rangle\tau\nabla\langle b^2\rangle + C_4\lambda^2\nabla\langle b^2\rangle\right] \tag{19}$$

$$\text{where } C_1 = \frac{7}{45},\quad C_2 = -\frac{203}{5400},\quad C_3 = \frac{403}{8100},\quad \text{and}\quad C_4 = -\frac{1}{6}$$

Here, λ is the characteristic length scale of turbulence. We now compute the curl of the mean velocity field $\overline{\mathbf{V}}$. In cylindrical coordinates, this reads:

$$\nabla\times\overline{\mathbf{V}} = \left(-r\frac{\partial\Omega}{\partial z} + \frac{1}{r}\frac{\partial V_z}{\partial\phi}\right)\hat{r} - \frac{\partial V_z}{\partial r}\hat{\phi} + (2-q)\Omega\hat{z} \tag{20}$$

Here, q is the shear parameter defined by:

$$q = -\frac{d\ln\Omega}{d\ln r} \tag{21}$$

Assuming axisymmetry and negligible vertical shear ($\partial\Omega/\partial z = 0$), the expression reduces to:

$$\nabla\times\overline{\mathbf{V}} = (2-q)\Omega\hat{z} \tag{22}$$

Now, after algebraic manipulations, we will get the final expression of the Vishniac flux for the turbulence profiles:

$$\nabla\cdot\boldsymbol{\mathcal{F}}^{NV} = \frac{2(2-q)R_\Omega\eta^2 z\xi}{qh^4}(AC_2 - C_1\xi - C_3 - C_4)$$

We introduced the dimensionless parameter R_Ω here, which will control Ω by the relation:

$$\Omega = \frac{-R_\Omega \eta}{qh^2} \tag{23}$$

Here, η_t is the turbulent diffusivity given by:

$$\eta_t = \frac{1}{3}\tau v^2 \tag{24}$$

3 The new induction equation

As given in equation 24, η_t is dependent on the form of turbulent velocity profile we adopted, thus in our system we have a stratified turbulent diffusivity. This means there will be an extra term in the induction equation, thus the mean field induction equation becomes:

$$\frac{\partial \overline{\mathbf{B}}}{\partial t} = \underbrace{\nabla \times \left[\overline{\mathbf{V}} \times \overline{\mathbf{B}} + \alpha\, \overline{\mathbf{B}}\right]}_{\text{Induction + } \alpha\text{-effect}} + \underbrace{\eta_t \nabla^2 \overline{\mathbf{B}}}_{\text{Turbulent diffusion}} \underbrace{-\left(\nabla \times \overline{\mathbf{B}}\right) \times \nabla \eta_t}_{\text{New stratification term}} \tag{25}$$

4 The model

4.1 Assumptions

We adopt a cylindrical coordinate system (r, ϕ, z) and solve the induction equation (Equation 3) under the assumption of axisymmetry, which neglects azimuthal (ϕ) derivatives. Applying the thin-disc approximation, the vertical magnetic field component is considered negligible compared to the radial and azimuthal components, i.e., $|\overline{B}_z| \ll |\overline{B}_r|, |\overline{B}_\phi|$, and radial ($r$) derivatives are neglected. These simplifications reduce the problem to a one-dimensional form in z, allowing local dynamo solutions. We solve for $\overline{B}_r$, $\overline{B}_\phi$, and α_m. Using $\nabla \cdot \overline{\mathbf{B}} = 0$, the vertical magnetic field $\overline{B_z}$ is determined.

The reduced induction equations for the $\alpha\Omega$ dynamo are:

$$\frac{\partial \overline{B}_r}{\partial t} = -\frac{\partial}{\partial z}\left(\alpha \overline{B}_\phi\right) + \eta_t \frac{\partial^2 \overline{B}_r}{\partial z^2} + \frac{\partial \eta_t}{\partial z}\frac{\partial \overline{B}_r}{\partial z^2} \tag{26}$$

$$\frac{\partial \overline{B}_\phi}{\partial t} = -\Omega \overline{B}_r + \eta_t \frac{\partial^2 \overline{B}_\phi}{\partial z^2} + \frac{\partial \eta_t}{\partial z}\frac{\partial \overline{B}_\phi}{\partial z^2} \tag{27}$$

The α_m evolution equation given in equation 8 becomes:

$$\frac{\partial \alpha_m}{\partial t} = -\frac{2\eta}{l^2 B_{\text{eq}}^2}\left[\alpha\left(\overline{B}_r^2 + \overline{B}_\phi^2\right) - \eta_t\left(\frac{\partial \overline{B}_r}{\partial z}\overline{B}_\phi - \frac{\partial \overline{B}_\phi}{\partial z}\overline{B}_r\right)\right] - \nabla \cdot \boldsymbol{\mathcal{F}}^{NV} - \nabla \cdot \boldsymbol{\mathcal{F}}^{D}, \tag{28}$$

The magnetic pitch angle is the angle between the azimuthal and radial components of the magnetic field in a disk, indicating the spiral structure of the field:

$$p = \arctan\left(\frac{\overline{B}_r}{\overline{B}_\phi}\right), \quad p \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \tag{29}$$

In a thin disc, the α-effect is written as:

$$\alpha_k = \alpha_0(r)\tilde{\alpha}(z), \tag{30}$$

where $\alpha_0 = \frac{l^2\Omega}{h}$ with h as the disc half-thickness. From symmetry considerations, α is an odd function of z, a conventional choice of the function is[4]:

$$\tilde{\alpha}(z) = \sin\left(\frac{\pi z}{h}\right). \tag{31}$$

We consider specific models at $r = 4\,\mathrm{kpc}$. Dimensionless control parameters are defined as:

$$R_\alpha \equiv \frac{\alpha_0 h}{\eta}, \quad R_\Omega \equiv \frac{-q\Omega h^2}{\eta} \tag{32}$$

Lengths are scaled to the disk half-thickness h, except for radial measurements, which are scaled to the disk radius R. The magnetic field is scaled to $B_0 \sim 1\,\mu\mathrm{G}$.

4.2 Numerical Methods

The equations (26), (27), and (28) are solved numerically using a simulation code in FORTRAN. The numerical schemes used in this project are inspired by the well-established Pencil Code, though the implementation itself has been developed independently from scratch.

4.3 Initial and Boundary Conditions

The choice of initial conditions is not critical in kinematic mean-field dynamo simulations, as the system tends to forget its initial state due to exponential growth (or decay) of magnetic modes. Here, a seed field with a Gaussian profile of amplitude 10^{-2} is used.

For boundary conditions, the standard vacuum boundary conditions are employed, where the region outside the dynamo-active disk is assumed to be a perfect electrical insulator.

$$\overline{B}_r = \overline{B}_\phi = 0, \quad \alpha_m = 0, \quad \text{at } z = \pm h. \tag{33}$$

Physically, this implies that no electric currents exist outside the disk, and hence, only the meridional components of the magnetic field (B_r and B_z) can penetrate the boundary, while the azimuthal component B_ϕ must vanish at the edge. At first glance, this assumption may seem questionable - after all, real galactic disks are enveloped by a halo of ionized gas which does conduct. However, this ionized corona is only weakly conducting compared to the dense interstellar medium in the disk, and it behaves like a turbulent electric vacuum for the purposes of modeling large-scale fields inside the disk.

4.4 Implementation

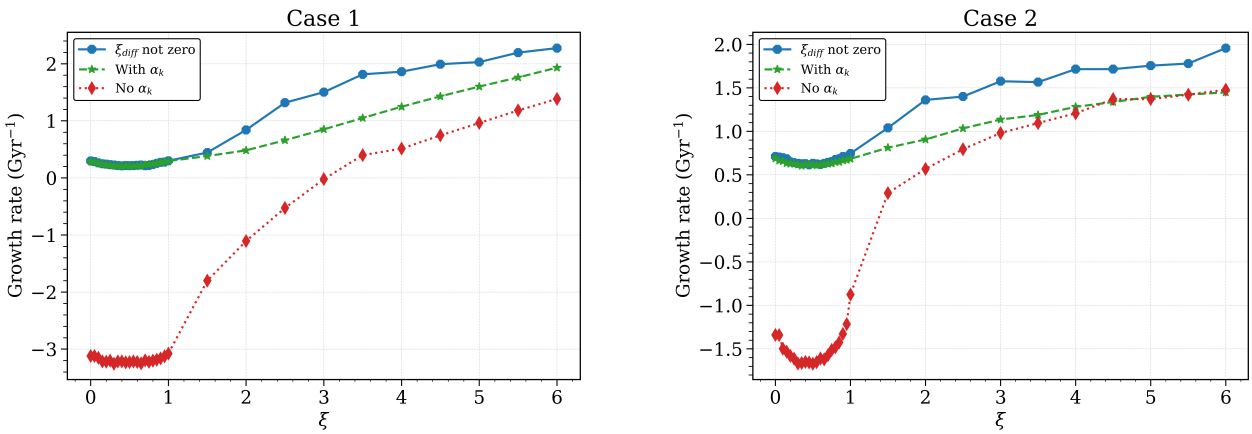
Spatial derivatives are approximated using a sixth-order finite differencing method, which balances accuracy and efficiency. Higher-order schemes (8th and 12th) showed only marginal gains in accuracy but were more computationally demanding, requiring increased memory and longer run times. Hence, the sixth-order method was chosen as the most efficient solution.

Time evolution is handled using a third-order implicit Runge–Kutta (RK3) scheme [3]. Although a fourth-order explicit RK4 scheme was also tested, no significant difference in the results was observed. A key advantage of the implicit RK3 over the explicit RK4 scheme is its stability when handling the stiff equations that arise in MHD simulations, where processes like diffusion and advection operate on different timescales. The implicit RK3 scheme enables larger time steps while maintaining stability, making it more computationally efficient than the RK4 scheme, which requires significantly smaller time steps, thereby increasing memory usage and run time to maintain stability.

The simulation employs a spatial resolution of 101 grid points. While models were also tested with 51 and 201 grid points, the 201-grid resolution, though providing slightly more accurate results, was significantly more time-consuming. Therefore, a resolution of 101 grid points was used, and every new model was also tested with 201 grid points to see if there are any pronounced differences.

Boundary conditions are implemented using ghost zones, which are additional grid points beyond the physical domain of the simulation. These ghost zones are not part of the actual solution, but are used to impose boundary conditions by mirroring, extrapolating, or applying specific rules to the data inside the physical grid. In the code, three ghost cells are placed on each side of the physical grid, resulting in a total of 107 grid points. The use of ghost zones is important for maintaining the accuracy of the finite differencing scheme near the boundaries, as they allow for consistent calculation of derivatives and prevent numerical instabilities that might arise from directly enforcing boundary conditions.

5 Results and Analysis



(a) Case 1, gaussian velocity profile.

(b) Case 2, inverted gaussian velocity profile.

Fig. 1: Growth rate values for Case 1 and Case 2 across different ξ_0 values. The models given in blue and green include the kinetic α-effect (α_k). In blue plot the diffusion is varied with ξ_0 and in green plot, it is kept a constant. The red plot have contsnt diffusion but kinetic α-effect is turned off. Case 1 and Case 2 correspond to the different velocity profiles given in 1

5.1 The Vishniac Flux in Mean-Field Dynamos: Aid or Hindrance?

We carried out a parameter sweep over ξ_0,from 0 to 6 in small steps. This was done to assess whether the Vishniac flux helps or hinders the dynamo action, and to check if there's a critical

window of ξ_0 values where it begins to help. The plots of growth rate vs ξ_0 and saturation strength vs ξ_0 for both Case 1 and Case 2 are shown in the figures.

We varied ξ_0 from 0.01 to 3.5. Although such large values of ξ_0 are not realistic in astrophysical settings, we included them intentionally to explore the theoretical behavior of the model. In reality, astrophysical ξ_0 is expected to be much lower, but it's at these large values that the Vishniac flux starts significantly boosting the dynamo action.

In Figure 1, we show how the growth rate changes as ξ_0 varies from 0.01 to 3.5. The red line shows the growth rate from the conventional model without the Vishniac term. Looking at Case 1, it's clear that below a certain critical ξ_0, the Vishniac flux works against the dynamo by reducing the growth rate. This is consistent with the derived expression for the Vishniac flux in Case 1:

$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} \propto \left[0.079 - 0.156\xi_0 e^{-2z^2/h^2}\right] \tag{34}$$

From this, we can see that there's a critical value of ξ_0 where the sign of the flux changes. This value can be calculated numerically and comes out to be $\xi_0 = 0.506$. In Figure 1a, we can clearly see the effect of this transition point. For $\xi_0 < 0.506$, the flux is negative and contributes negatively to the evolution of α_m. It essentially fights against the driving term provided by α_k. In simulations without α_k and for $\xi_0 < 0.506$, the magnetic field decays, and the decay rate slows as ξ_0 approaches 0.506. This trend matches what we expect from the midplane analysis.

Once $\xi_0 > 0.506$, the flux flips sign and starts to help the dynamo. This is also seen in the sharp increase in growth rate past $\xi_0 = 0.5$ in Figure 1a. For $\xi_0 > 1$, where the small-scale fields become super equipartition and start dominating, the flux enhances the growth rate beyond that of the conventional dynamo. This suggests that at large enough ξ_0, the Vishniac term could possibly dominate the helicity evolution and drive the dynamo by itself. This is discussed more in the next section.

Now coming to Case 2, the flux takes the form:

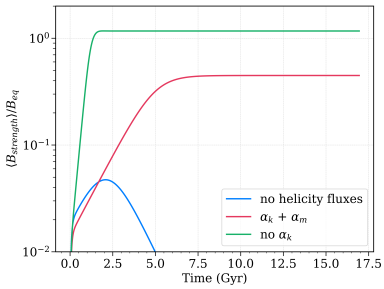
$$-\nabla \cdot \boldsymbol{\mathcal{F}}^{NV} \propto \left[0.154 - 0.156\xi_0\right] e^{-2z^2/h^2} \tag{35}$$

Here, the saddle point where the flux flips sign is at $\xi_0 = 0.987$. There is also an additional exponential factor outside the bracket in Case 2, which makes the flux look like a bell curve peaking at the mid plane. This means the Vishniac term has a very localized effect in this case, most of the action happens near the mid plane.

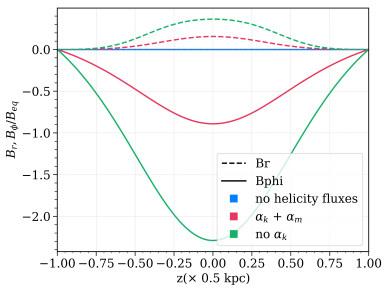
From simulations (see Figure 1b), we observe that the overall trend in Case 2 is similar to Case 1. However, the increase in growth rate with increasing ξ_0 is slower. Even at $\xi_0 = 3.5$, the growth rate has not yet surpassed that of the conventional dynamo. This tells us that unless ξ_0 becomes very large, the flux remains too localized to be significantly helpful. For it to become effective, the value of ξ_0 must be high enough for the flux's influence to extend beyond just the midplane.

5.2 The Vishniac flux action without α_k driving the dynamo

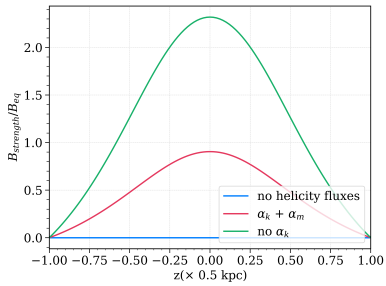
As shown in the previous section, there is evidence that at super equipartition values of ξ_0, the Vishniac flux alone can sustain dynamo action. To investigate this possibility, we considered three models. One without any helicity fluxes, only α_k, then one with α_k and helicity fluxes, and then a model with only the Vishniac effect, no α_k. Surprisingly, in the case with α_k turned



(a) Mean field strength vs time plots, the y axis here is in logarithmic scale.



(b) Mean field strength vs z plots, dashed line represents $\overline{B}_\phi$ and solid lines are $\overline{B}_r$.



(c) Total field strength vs z plots. This is the vector sum of $\overline{B}_\phi$ and $\overline{B}_r$

.

Fig. 2: In the blue plots, there is no magnetic helicity fluxes included, thus it is catastrophically decaying. In the red plot, there is a diffusive flux included, and both blue and red models have α_k in it. But in the green model, there is a dynamo action purely from Vishniac flux.

off and ξ_0 set to 3.5, we still observed a growing and eventually saturated magnetic field—driven purely by the Vishniac flux.

But how is this possible, given that $\alpha_k = 0$? In the conventional picture of a galactic dynamo, turbulence (e.g., from supernova explosions) produces kinetic helicity, breaking reflectional symmetry and giving rise to α_k. This, in combination with differential rotation (Ω), forms the classic $\alpha\Omega$ dynamo loop: poloidal fields generate toroidal fields and vice versa, leading to exponential field growth. When α_k is switched off, that feedback loop is broken, and we expect no sustained dynamo.

However, the Vishniac flux introduces a twist to this story. In the standard scenario, the role of the flux is to avoid catastrophic α-quenching by transporting small-scale magnetic helicity out of the dynamo-active region. But at sufficiently large ξ_0, this flux doesn't only work to remove excess helicity, but also behave as the driving term of magnetic helicity evolution equation.

This can be explained using the evolution equation for magnetic α, assume a high ξ_0 case where the flux has an overall positive sign and is larger than diffusion:

$$\frac{\partial \alpha_m}{\partial t} \approx -\frac{2\eta}{l^2 B_{\rm eq}^2}\left[\alpha\left(\overline{B}_r^2 + \overline{B}_\phi^2\right) - \eta\left(\frac{\partial \overline{B}_r}{\partial z}\overline{B}_\phi - \frac{\partial \overline{B}_\phi}{\partial z}\overline{B}_r\right)\right] + \boldsymbol{\mathcal{F}}^{\rm NV}. \tag{36}$$

In the early stages, when mean fields are still weak, the non-linear quenching and gradient terms are negligible. Thus, the equation 8 will not be activated yet, and α_k and Ω drive the equations 26 and 27. Once the mean fields start growing, the equation 8 begins to affect the system. Now, no α_k means no mean field. However, since the Vishniac flux can exist without any mean fields, the equation 8 will still work to evolve the system from the initial growth phase. Thus, at a large and positive Vishniac flux, the term $\boldsymbol{\mathcal{F}}^{\rm NV}$ directly activates α_m. That is, the helicity transport creates a net gradient in magnetic helicity, mimicking the role of α_k. Therefore, the Vishniac flux-generated α_m results in a non-zero total α-effect, which, together with Ω, forms the positive feedback loop required for classic mean-field dynamo action.
In short, a sufficiently strong Vishniac flux seeds an effective α-effect which is completely magnetic in origin, which enables the classic dynamo feedback loop without needing kinetic helicity. This means a dynamo is possible that is purely driven by magnetic energy.

6 Conclusions

In this study, we introduced the Vishniac flux into numerical dynamo models to assess its role in the evolution of galactic magnetic fields. Our primary question was whether this new flux could significantly alter our understanding of mean-field dynamo theory. The short answer is, not by much. The Vishniac flux alone does not independently drive a dynamo to saturation for realistic values of ξ_0.

The Vishniac flux can theoretically drive dynamo action on its own at very high values of ξ_0, due to its quadratic dependence on this parameter. However, such conditions requiring super equipartition magnetic fields are unlikely in realistic galactic environments. Thus, while the flux introduces a magnetically driven dynamo mechanism, it does not independently sustain a dynamo under realistic conditions.

When combined with a conventional α_k-driven dynamo, the Vishniac flux can either suppress or enhance magnetic field growth, depending on the value of ξ_0 and the turbulence profile. The flux reverses sign at a critical ξ_0, which varies with turbulence shape: around 0.506 for edge-peaked turbulence and around 1 for midplane-peaked turbulence. Below this threshold, it hinders dynamo action; above it, it supports it.

A major limitation of the model studied here is the assumption of a vertically constant density profile. This is unrealistic, although we did investigate different turbulent profiles and found that the underlying effect of the Vishniac flux remained qualitatively the same. This is an area where the model can be improved. Another obvious direction for improvement is to relax the thin-disc approximation. Incorporating the galactic corona and solving the system in two dimensions could yield global dynamo solutions and potentially offer deeper insights into the underlying physics.

In summary, this project has contributed to our understanding of galactic dynamo theory by incorporating and thoroughly analyzing the Vishniac flux within a dynamical quenching framework. The discovery that a purely magnetically driven dynamo may be possible introduces promising new directions for both theoretical and observational research. Future studies that expand on this work using more realistic galactic models and multidimensional simulations are likely to provide a deeper understanding of the complex processes that maintain cosmic magnetic fields.

Bibliography

[1] Rainer Beck, Luke Chamandy, Ed Elson, and Eric G. Blackman. Synthesizing observations and theory to understand galactic magnetic fields: Progress and challenges, 2019. URL: https://arxiv.org/abs/1912.08962, arXiv:1912.08962.

[2] Eric G. Blackman. Magnetic helicity and large scale magnetic fields: A primer. *Space Science Reviews*, 188(1–4):59–91, April 2014. URL: http://dx.doi.org/10.1007/s11214-014-0038-6, doi:10.1007/s11214-014-0038-6.

[3] Axel Brandenburg. *Computational aspects of astrophysical MHD and turbulence*, page 269–344. CRC Press, April 2003. URL: http://dx.doi.org/10.1201/9780203493137.ch9, doi:10.1201/9780203493137.ch9.

[4] Luke Chamandy, Anvar Shukurov, Kandaswamy Subramanian, and Katherine Stoker. Non-linear galactic dynamos: a toolbox. *Monthly Notices of the Royal Astronomical Society*, 443(3):1867–1880, 07 2014. arXiv:https://academic.oup.com/mnras/article-pdf/443/3/1867/4910392/stu1274.pdf, doi:10.1093/mnras/stu1274.

[5] Frederick A. Gent, Mordecai-Mark Mac Low, and Maarit J. Korpi-Lagg. Transition from Small-scale to Large-scale Dynamo in a Supernova-driven, Multiphase Medium. *The Astrophysical Journal*, 961(1):7, 1 2024. doi:10.3847/1538-4357/ad0da0.

[6] Kishore Gopalakrishnan and Kandaswamy Subramanian. Magnetic helicity fluxes from triple correlators. *The Astrophysical Journal*, 943(1):66, jan 2023. URL: https://dx.doi.org/10.3847/1538-4357/aca808, doi:10.3847/1538-4357/aca808.

[7] Oliver Gressel. Supernova-driven turbulence and magnetic field amplification in disk galaxies, 2009. URL: http://nbn-resolving.de/urn:nbn:de:kobv:517-opus-29094.

[8] Anvar Shukurov and Kandaswamy Subramanian. *Astrophysical Magnetic Fields: From Galaxies to the Early Universe*. Cambridge Astrophysics. Cambridge University Press, 2021.

[9] S. Sur, A. Shukurov, and K. Subramanian. Galactic dynamos supported by magnetic helicity fluxes. , 377:874–882, May 2007. arXiv:arXiv:astro-ph/0612756, doi:10.1111/j.1365-2966.2007.11662.x.

[10] E. T. Vishniac. A Simple Model for the Galactic Dynamo. In *American Astronomical Society Meeting Abstracts #220*, volume 220 of *American Astronomical Society Meeting Abstracts*, page 308.05, May 2012. URL: http://adsabs.harvard.edu/abs/2012AAS...22030805V.

[11] E. T. Vishniac and J. Cho. Magnetic Helicity Conservation and Astrophysical Dynamos. , 550:752–760, April 2001. arXiv:arXiv:astro-ph/0010373, doi:10.1086/319817.